Digital Logic Circuits (Spring 2026)
Quiz #5

Name: ____________________ Student ID: ____________________

1. How the keyword "TYPE" is used? b

a) TYPE datatype_name IS type_from_predefined_datatypes;
b) TYPE datatype_name IS datatype_range;
c) TYPE datatype_range IS datatype_name;
d) USE TYPE datatype_range IS datatype_name;

2. What does the architecture of an entity define? b

a) External interface
b) Internal functionality
c) Ports of the entity
d) Specifications

3. The process can be ___________ by using WAIT statements. c

a) Suspended
b) Resumed
c) Suspended as well as resumed
d) Cannot be determined

4. Which of the following is a wrong declaration for a new data type? d

a) TYPE my_logic IS RANGE 0 to 100;
b) b. TYPE my_logic IS ('0', '1', '2');
c) c. TYPE my_logic IS ARRAY (0 TO 3) OF BIT;
d) d. TYPE my_logic IS <0 TO 20 >

5. For any concurrent assignment statement, which of the following is true? d

a) The statement is executed once
b) The statement is executed twice
c) The value of the left operand is assigned to the right operand
d) The statement is executed as many times as the value changes

6. Which of the following is the correct use of the signal? b

a) To set a default value
b) To pass value between circuits
c) To declare a variable
d) To represent local information

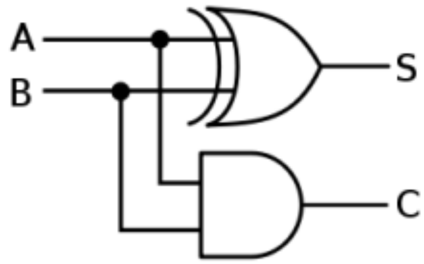
7. Which of the following is not a characteristic of combinational circuits? c

a) The output of a combinational circuit depends on the present input
b) There is no use of clock signals in combinational circuits
c) The output of the combinational circuit depends on the previous output
d) There is no storage element in a combinational circuit

8. **Which of the following is not a combinational circuit?** d

a) Adder
b) Code converter
c) Multiplexer
d) Counter

9. **Write a VHDL process that describes the digital circuit shown below. What is the main purpose/function of this circuit?**




```
p_adder : process(A, B)
begin
      S <= A xor B    ;
      C <= A and B    ;
end process;
```

This is a half-adder with carry. It adds to bits *A* and *B* and has two outputs *Sum(S)* and *Carry(C)*. The carry signal represents an overflow.

10. **Please fill in the following TWO empty boxes in the VHDL description for a D-flip flop (register) with an active high enable signal and no reset signal.**

```
p_ff : process(clk)
begin
   if rising_edge(  clk ) then
     If enable = '1'  then
          Q <= D           ;
     end if;
   end if;
end process;
```